

Python Interview Questions

Day 02 • Friends Version

1. What is the difference between / and // in Python?

Answer: The / operator performs regular division and returns a floating-point result, while // performs floor division and returns the quotient rounded down.

Example:

```
print(7 / 2)
print(7 // 2)
```

Output:

```
3.5
3
```

2. What is the difference between break and continue?

Answer: break immediately terminates a loop, while continue skips the remaining statements in the current iteration and moves to the next iteration.

Example:

```
for i in range(5):
    if i == 2:
        continue
    print(i)
```

Output:

```
0
1
3
4
```

3. What is the purpose of the range() function?

Answer: range() generates a sequence of numbers, commonly used for iteration in loops. It can accept start, stop, and step values.

Example:

```
for i in range(1, 4):
    print(i)
```

Output:

```
1
2
3
```

4. What is the difference between a for loop and a while loop?

Answer: A for loop is commonly used to iterate over an iterable, while a while loop repeatedly executes a block as long as a condition is true.

Example:

```
for i in range(3):
    print(i)
```

Output:

0
1
2

5. What is the purpose of the len() function?

Answer: The len() function returns the number of items in an object such as a string, list, tuple, set, or dictionary.

Example:

```
name = "Python"  
print(len(name))
```

Output:

6

5 Questions Completed